

Forewarned is Forearmed: A Survey on Large Language Model-based Agents in Autonomous Cyberattacks

MINRUI XU, Nanyang Technological University, Singapore

JIANI FAN, Nanyang Technological University, Singapore

XINYU HUANG, University of Waterloo, Canada

CONGHAO ZHOU, University of Waterloo, Canada

JIAWEN KANG, Guangdong University of Technology, China

DUSIT NIYATO, Nanyang Technological University, Singapore

SHIWEN MAO, Auburn University, USA

ZHU HAN, University of Houston, USA

XUEMIN (SHERMAN) SHEN, University of Waterloo, Canada

KWOK-YAN LAM, Nanyang Technological University, Singapore

With the continuous evolution of Large Language Models (LLMs), LLM-based agents have advanced beyond passive chatbots to become autonomous cyber entities capable of performing complex tasks, including web browsing, malicious code and deceptive content generation, and decision-making. By significantly reducing the time, expertise, and resources, AI-assisted cyberattacks orchestrated by LLM-based agents have led to a phenomenon termed Cyber Threat Inflation, characterized by a significant reduction in attack costs and a tremendous increase in attack scale. To provide actionable defensive insights, in this survey, we focus on the potential cyber threats posed by LLM-based agents across diverse network systems. Firstly, we present the capabilities of LLM-based cyberattack agents, which include executing autonomous attack strategies, comprising scouting, memory, reasoning, and action, and facilitating collaborative operations with other agents or human operators. Building on these capabilities, we examine common cyberattacks initiated by LLM-based agents and compare their effectiveness across different types of networks, including static, mobile, and infrastructure-free paradigms. Moreover, we analyze threat bottlenecks of LLM-based agents across different network infrastructures and review their defense methods. Due to operational imbalances, existing defense methods are inadequate against autonomous cyberattacks. Finally, we outline future research directions and potential defensive strategies for legacy network systems.

CCS Concepts: • **Networks** → *Network security*; • **Computing methodologies** → *Artificial intelligence*; • **General and reference** → *Surveys and overviews*.

Additional Key Words and Phrases: Large Language Models (LLMs), Cybersecurity, Autonomous Cyberattacks, Network Security.

1 Introduction

1.1 Background and Motivation

The evolving capabilities of large language models (LLMs) are rapidly transforming attack and defense operations in cybersecurity [80]. Major AI companies have begun to systematically evaluate these risks using the Cyber Kill Chain Framework [127, 161]. For instance, Google’s Project Naptime team has demonstrated that frontier LLMs can

Authors’ Contact Information: Minrui Xu, MINRUI001@e.ntu.edu.sg, Nanyang Technological University, Singapore; Jiani Fan, jiani001@e.ntu.edu.sg, Nanyang Technological University, Singapore; Xinyu Huang, x357huan@uwaterloo.ca, University of Waterloo, Waterloo, ON, Canada; Conghao Zhou, c89zhou@uwaterloo.ca, University of Waterloo, Waterloo, ON, Canada; Jiawen Kang, kavinkang@gdut.edu.cn, Guangdong University of Technology, Guangzhou, Guangdong, China; Dusit Niyato, dnyato@ntu.edu.sg, Nanyang Technological University, Singapore; Shiwen Mao, smao@ieee.org, Auburn University, USA; Zhu Han, hanzhu22@gmail.com, University of Houston, Houston, TX, USA; Xuemin (Sherman) Shen, sshen@uwaterloo.ca, University of Waterloo, Waterloo, ON, Canada; Kwok-Yan Lam, kwokyan.lam@ntu.edu.sg, Nanyang Technological University, Singapore.

autonomously assist in offensive security tasks with minimal human input, including code exploitation and vulnerability discovery [75]. Similarly, Anthropic has deployed red teams, i.e., offensive attackers, to test its Claude models against cybersecurity misuse scenarios, revealing new emergent risks in autonomous agent behavior [23]. These findings reinforce the concern that LLMs have significantly lowered the technical threshold and cost of multi-stage intrusions [175]. Leveraging LLMs equipped with perception, memory, reasoning, and action modules, LLM-based agents can conduct cyberattacks autonomously with minimal human intervention [47, 107]. Specifically, LLM-based agents introduce novel attack paradigms, e.g., jailbreak attack [170], and significantly amplify existing cyberattacks, e.g., vulnerability exploitation, malware generation, and social engineering [38]. LLM-based agents allow attackers with limited skills and resources to conduct complex cyberattacks with minimal human intervention. Therefore, LLM-based agents accelerate attack deployment, scale offensive activities, and erode traditional resource bottlenecks, resulting in Cyber Threat Inflation. This inflation in cybersecurity describes the drastic reduction in operational costs for launching cyberattacks alongside a significant increase in their scalability.

LLM-based agents can reduce time, expertise, and resource requirements across all stages of cyberattacks, e.g., vulnerability detection, customized exploitation, and persistent installation [161]. Therefore, cyberattacks that previously required months of labor and substantial expert involvement can now be accomplished within hours [157]. In addition to cost collapse, scale uplift manifests in three critical dimensions [18]. First, capability uplift refers to the automation of offensive tasks such as vulnerability scanning and social engineering, traditionally limited to skilled red-team experts. For instance, PentestGPT [52] demonstrates a 228.6% increase in task completion, and RapidPen [132] achieves shell access within 200–400 seconds at an estimated cost of \$0.3–\$0.6 per run, with a 60% success rate. Second, throughput uplift captures the ability of LLM-based agents to execute continuous and large-scale attacks in parallel. To generate next-packet predictions based on previous traffic context in unmanned aerial vehicle (UAV) networks, Net-GPT [151] achieves 95% packet-generation accuracy and maintains time man-in-the-middle (MitM) sessions for 30 min without expert intervention. Finally, autonomous risk emergence highlights how LLMs with reasoning abilities can dynamically adapt to defensive mechanisms. In satellite networks, PLLM-CS [85] autonomously interprets satellite telemetry to detect intent-based anomalies, signaling the rise of real-time, self-adjusting adversarial agents.

While advanced persistent threat (APT) groups often leverage sophisticated techniques such as advanced phishing [42], zero-day exploitation [222], and polymorphic malware [162], individual attackers have also demonstrated the ability to execute similar methods. However, the emergence of LLM-based agents will empower individual attackers to achieve sophisticated attacks. Through the integration of LLMs with tool APIs and accessible programming interfaces, organizations with limited technical capabilities are now able to orchestrate complex operations, encompassing systematic vulnerability assessment and coordinated exploit deployment. This transformation has effectively dismantled the traditional security asymmetry between attackers and defenders, as sophisticated attack vectors are no longer restricted to well-resourced threat actors. Furthermore, an LLM-based agent might probe systems outside of typical human working hours or adapt in real-time. Therefore, defenses should remain vigilant at all times to detect and respond to these autonomous intrusions. Consequently, the potential for widespread, cost-effective system compromises has expanded dramatically.

The cyber threat inflation has profound implications for legacy network infrastructures, including enterprise networks, cellular core networks, cloud platforms, and embedded systems. However, many applications of LLMs in cybersecurity still assume traditional threat models, where human attackers remain the principal adversaries [144]. While human attackers have traditionally posed the primary threat in cybersecurity, integrating LLMs now augments or even substitutes their expertise in many domains. Accordingly, LLMs are employed to enhance performance across established cybersecurity tasks and evaluation benchmarks. For instance, LLMCloudHunter [164] and AppPoet [218] present targeted solutions

Table 1. Related works on LLM Agents, cyberattacks, and network systems.

Ref.	Survey Focus	LLM agents	Cyber-attacks	Networks
[189]	Architecture, capabilities, applications, and evaluation of LLM-based agents	✓	✗	✗
[123]	The life-cycle of LLM agents including construction, collaboration, and evolution	✓	✗	✗
[97]	LLM applications in software engineering and evolution into agents	✓	✗	✗
[86]	LLM-based multi-agent systems for software engineering and human-in-the-loop	✓	✗	✗
[214]	LLMs for cybersecurity tasks like threat intelligence and vulnerability detection	✗	✓	✗
[65]	Benchmarking 42 LLMs on intrusion and malware detection tasks	✗	✓	✗
[221]	Evaluation of 37 LLMs for bug detection and patch generation	✗	✓	✗
[27]	LLMs for code security, strengths in simple flaws and weaknesses in complex issues	✗	✓	✗
[80]	Frontier AI's impact on cybersecurity landscapes	✗	✓	✗
[11]	LLMs for malware detection, task taxonomies, metrics, and countermeasure	✗	✓	✗
[95]	LLM usage in code analysis, malware detection, and reverse engineering	✗	✓	✗
[135]	LLM-specific threats and defense pipelines in 6G networks	✗	✓	✓
[58]	Cyberattacks on cyber-physical systems; threat modeling and defense synthesis	✗	✓	✓
[37]	ML-enabled attacks on IoT networks; evaluation challenges and defense gaps	✗	✓	✓
[193]	Metaverse fundamentals, emerging security threats, and privacy challenges	✗	✓	✓
Ours	Cyberattack capabilities of LLM-based agents across various network systems	✓	✓	✓

for cloud threat intelligence extraction and Android malware detection, respectively, yet lack a systematic analysis for LLM-based cyberattack agents across different types of networks.

1.2 Related Works

As summarized in Table 1, the capabilities of LLM-based agents have expanded from simple chatbots to sophisticated copilots in cybersecurity, although their deployment across diverse network environments is still under investigation. From an architectural perspective, Wang *et al.* [189] provide a comprehensive review of LLM-based autonomous agents, focusing on their construction, capabilities, applications, and evaluation. Adopting a life cycle perspective, Luo *et al.* [123] categorize LLM-based agents into three dimensions, i.e., construction, collaboration, and evolution, encompassing components from profile definition to deployment in real-world settings. With a domain-specific focus, Jin *et al.* [97] review LLM applications in software engineering across six domains, examine their evolution into autonomous agents, and propose a taxonomy of agent capabilities. Considering multi-agent systems, He *et al.* [86] investigate LLM-based multi-agent systems in software engineering, outlining a two-phase research agenda while emphasizing the critical role of human-in-the-loop approaches for system advancement.

LLM adaptation and evaluation for cybersecurity applications have recently been mapped out in several complementary surveys. From a methodological standpoint, Zhang *et al.* [214] survey adaptation techniques that repurpose foundation models for threat intelligence and vulnerability detection, and articulate open challenges and integration requirements. From an empirical perspective, Ferrag *et al.* [65] benchmark 42 LLMs across intrusion and malware-detection tasks. In vulnerability remediation, Zhou *et al.* [221] assess 37 LLMs on bug detection and patch generation, advocating for

higher-quality datasets and tighter workflow integration. At the code level, Basic *et al.* [27] show that while LLMs handle simple flaws well, they still struggle with complex security issues. Adopting a policy lens, Guo *et al.* [80] analyze frontier AI's security impact through case studies and controlled evaluations, provide recommendations for policymakers, and identify pressing research gaps. Focusing on malware, Al *et al.* [11] summarize core concepts, task taxonomies, and evaluation metrics, and propose a risk-mitigation framework that balances theory with practice. Finally, Jelodar *et al.* [95] review LLM-enabled code analyses for malware detection, compare fine-tuning strategies, and highlight challenges.

Security risks and defenses for network systems, from 6G to cyber-physical infrastructures and the metaverse, have been scrutinized in recent research. For 6G, Nguyen *et al.* [135] survey LLM threats, vulnerabilities, mitigation pipelines, and blockchain integration for secure deployment. For cyber-physical systems, Duo *et al.* [58] analyze attacks and defense techniques. For IoT, Bout *et al.* [37] examine ML-enabled attacks, their advantages, and defense gaps. For metaverse, Wang *et al.* [193] map security threats and propose safeguards across human, physical, and digital realms. Different from these surveys, we provide a network-centric review examining LLM-based agents, their cyberattack capabilities, and impact across network paradigms. By understanding the core challenges of LLM-based agents like hallucination and context windows, defenders can identify weaknesses.

1.3 Contributions

Conventional perspectives in cybersecurity often overlook that LLM-based autonomous agents can be both defenders and adversaries, contributing to Cyber Threat Inflation to legacy systems [161]. This oversight reveals a gap in current cybersecurity research. To fill this gap, we provide a comprehensive taxonomy and comparative analysis of LLM-based agents in autonomous cyberattacks. We emphasize that LLM-based agents are not just tools for defenders but can become the adversaries themselves. Blue teams, i.e., defensive protectors, defending against cyberattacks, should update threat models by considering LLM-based agents as potential attackers and recognizing novel threat dynamics. We categorize research across attack chains and examine manifestations in static, mobile, and infrastructure-free networks. Existing analysis shows how LLM-based agents reduce attack costs while creating new defense challenges through automation and operational asymmetries.

In this survey, we investigate LLM-based autonomous agents as cyber adversaries operating autonomously. We begin by decomposing each LLM-based agent into five fundamental modules, consisting of models, perception, memory, reasoning & planning, and actions. Then, we demonstrate how multiple agents can collaborate with humans and other agents to deliver end-to-end attacks autonomously. Additionally, we examine how cyberattacks become more cost-effective and scalable while generating new forms of autonomous risk across network systems with diverse infrastructure characteristics. By highlighting classic defense methods that struggle against LLM-driven attacks, the paper provides advice for blue teams on where to watch out. The main contributions of this survey can be summarized as follows.

- (1) We present a novel unified architecture that abstracts the common design patterns of existing LLM-based cyberattack agents. This architecture comprises components for model selection, perception, memory, reasoning&planning, and tools&actions. We demonstrate that cooperative multi-agent orchestration can enable autonomous cyber operations.
- (2) We present a taxonomy of eight representative cyberattack capabilities for LLM-based agents, and analyze the specific attack bottlenecks and limitations these agents face in executing each of these capabilities.
- (3) We demonstrate how the cyberattack capabilities of LLM-based agents manifest across different network paradigms, including static infrastructure networks, mobile infrastructure networks, and infrastructure-free networks.

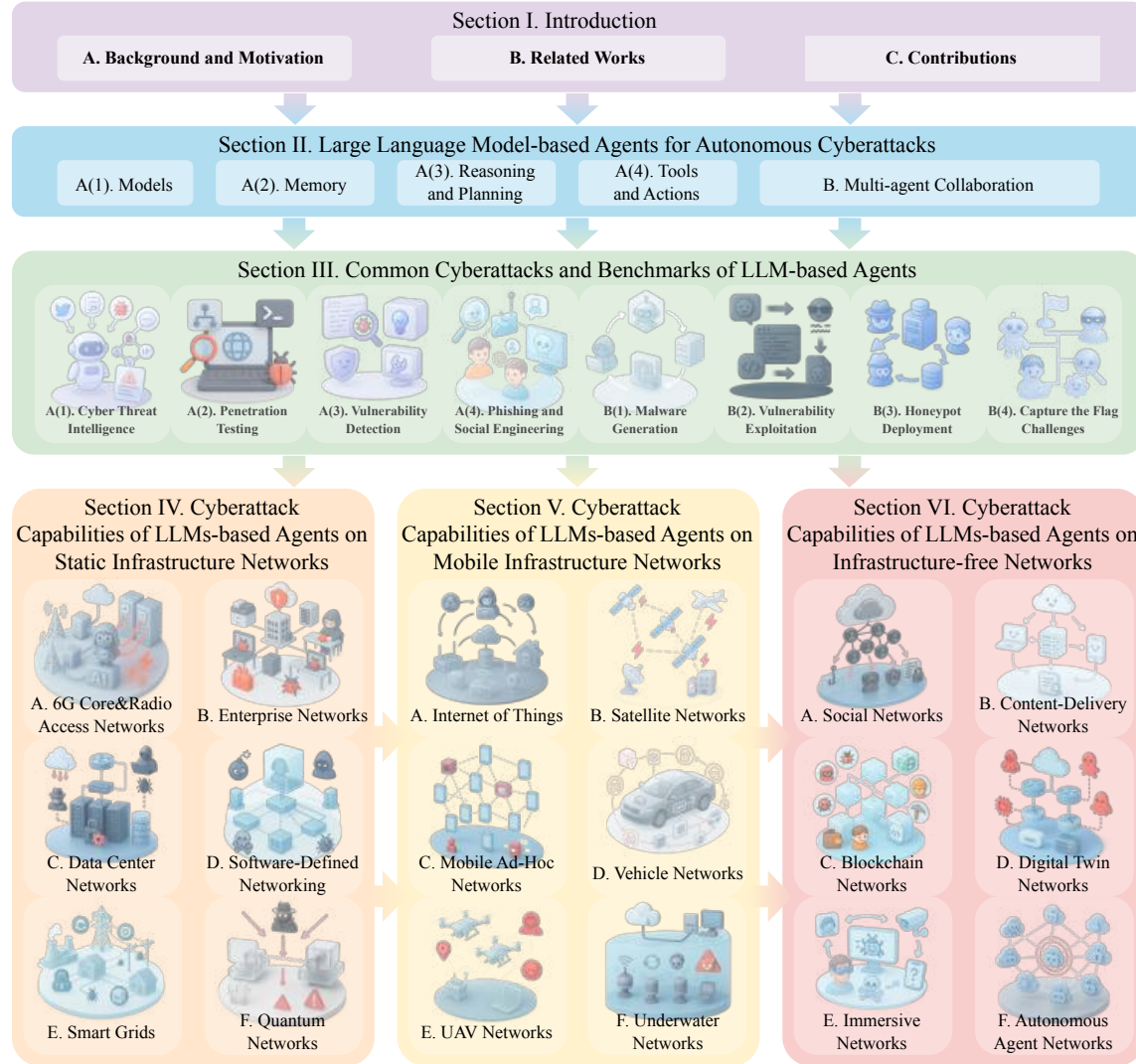


Fig. 1. The outline of this paper.

Section II deconstructs the construction and collaboration of LLM-based cyberattack agents. Section III presents common cyberattack capabilities of LLM-based agents and benchmarks. Sections IV, V, and VI then analyze how those capabilities manifest in three network paradigms, including static infrastructure networks, mobile infrastructure networks, and infrastructure-free networks, respectively. An overview of this survey's structure is shown in Fig. 1. With this survey, we provide a clear direction of how LLM-enabled adversaries evolve across capabilities and network systems. The analysis serves as a reference for blue-team defenders preparing defenses to track the state-of-the-art adversaries.